CG	CLASS INDEX NO.
07S ___	

CANDIDATE NAME: ________________________________

TAMPINES JUNIOR COLLEGE
PRELIMINARY EXAMINATION 2008

BIOLOGY 9747/03
Paper 3 Applications Paper

TUESDAY 19 AUGUST 2008 **2.30 – 4.00 PM (1 hour 30 minutes)**

Additional materials: Answer paper

INSTRUCTIONS TO CANDIDATES

1. Write your name, Civics group and candidate number in the spaces at the top of this page and on all separate answer paper used.
2. Write in dark blue or black pen on both sides of the paper.
3. You may use a soft pencil for any diagrams, graphs or rough working.
4. Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Write the answers to Questions 1, 2 and 3 in the spaces provided on the question paper. Answer Question 4 on the separate answer paper provided.
The answer to Question 4 should be illustrated by large, clearly labelled diagrams, where appropriate.

At the end of the examination:
Fasten all separate answer paper securely separate from the question paper.

INFORMATION FOR CANDIDATES
The number of marks is given in brackets [ ] at the end of
Each question or part question.

FOR EXAMINER'S USE	
1	
2	
3	
4	
TOTAL	

Answer **all** questions.

QUESTION ONE

After nearly a decade of setbacks and false starts, stem-cell science finally seems to be hitting its stride. Just a year after Japanese scientists first reported that they had generated stem cells by reprogramming adult skin cells — without using embryos — American researchers have managed to use that groundbreaking technique to achieve another scientific milestone. They created the first nerve cells from reprogrammed stem cells — an important demonstration of the potential power of stem-cell-based treatments to cure disease.

Led by Kevin Eggan at the Harvard Stem Cell Institute and Christopher Henderson at Columbia University, the 13-person team reported online today in *Science Express* that they had generated motor neurons from the skin cells of two elderly patients with a rare form of ALS, or Lou Gehrig's disease, a progressive neurodegenerative condition. The new study marks an important first step on the road toward real stem-cell-based therapies, and also answers several plaguing questions about the pioneering stem-cell technique known as induced pluripotent stem cell, or iPS, generation.

IPS was first described by Japanese biologist Shinya Yamanaka, who, in 2007, showed that the introduction of four genes into an adult human skin cell could reprogram it back to an embryonic state. Like embryonic stem cells, these reprogrammed adult cells could be coaxed into becoming any other type of cell — from skin to nerve to muscle.

Adapted from article from time.com

(a)(i) Embryonic stem cells can be isolated for research. State the location from which they can be isolated.

[1]

(ii) Describe the unique properties of stem cells that make them useful in research.

[2]

(iii) The passage mentioned 'induced pluripotent stem cells' (IPS)/reprogrammed stem cells that can be induced to form nerve cells to treat neurodegenerative diseases. Describe how this can be done *in vitro.*

____

____

____ [3]

(iv) During reprogramming in adult cells to form reprogrammed stem cells, some genes must be turned on. Suggest a gene that must be turned on during this process.

____ [1]

(b) Plant tissue culture is a method used to propagate plants. The flow diagram shows one method of plant tissue culture.

A meristematic explant is removed from a plant

↓

Using sterile conditions the explant is transferred to a vessel of culture containing nutrients.

↓

Callus tissue develops

↓

Callus tissue is transferred to a new culture medium containing nutrients and growth regulators

↓

New shoots or roots develop

↓

Developing plants are separated and grown under optimal conditions

(i) What is a callus?

____

____ [2]

(ii) What is a 'meristematic explant' and give one advantage in using an explant from the meristematic region for tissue culture?

____

____

____ [2]

(iii) Tissue culture is often used to propagate many genetically identical plants from the parent plant.
Describe briefly the process of this propagation with the use of callus tissue that has been isolated from the parent plant.

____

____

____ [2]

(iv) Another type of plant cell culture is the culture of protoplasts from plants.
If I want to combine the good traits of two related plant species together, protoplast fusion would be a more suitable method.
Outline briefly the process of protoplast fusion.

____

____

____ [2]

Total: 15 marks

QUESTION TWO

Severe Combined Immunodeficiency disease (SCID) is a genetic disease that involves no production of immuno-competent T and B lymphocytes, hence compromising the immune system of the patient. Opportunistic infections can kill the SCID patients before the first birthday.

Two types of SCID namely X-linked SCID and ADA SCID are the most common forms of SCID.

Figure 2.1 shows the inheritance of the disease in a family in which both parents are carriers.

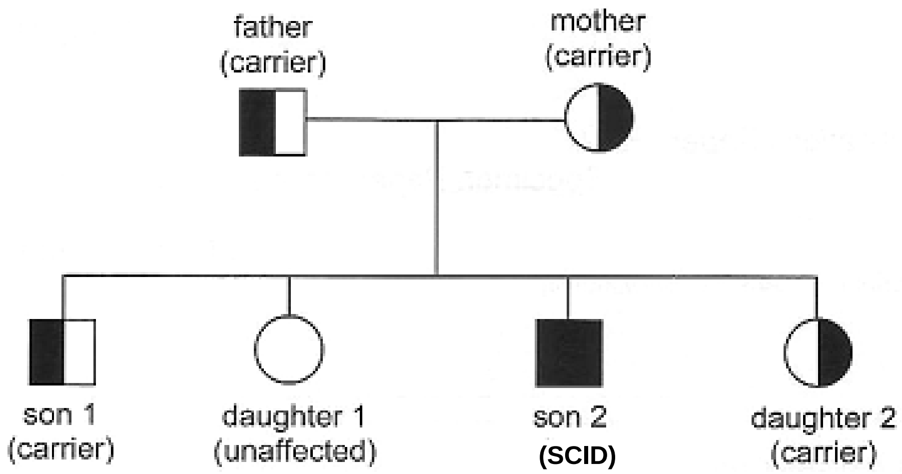


Fig. 2.1

(a)(i) State the form of SCID that is being inherited in the family.

____ [1]

(ii) Explain your answer in (a)(i) using the diagram in Fig. 2.1.

____

____ [2]

(iii) Describe briefly the genetic basis of X-linked SCID.

____

____

____

____

[4]

(b) 2 RFLP morphs have been found tightly linked to the SCID gene on chromosome 20 which can be used for genetic screening. A radioactive probe has been designed to reveal the RFLP #1 and RFLP #2.

The arrows on Fig. 2.1 show the position of the restriction sites of *EcoRI*. The asterisk (*) indicate the position of mutation at the restriction site that result in the loss of the restriction site.

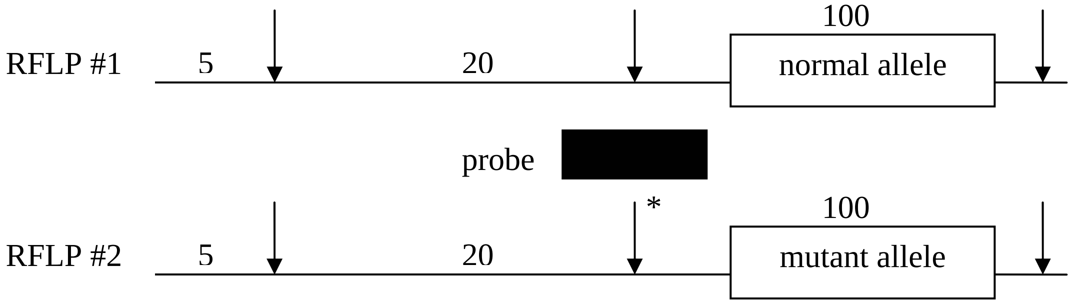


Fig. 2.1

.
In the space provided below, draw the resulting autoradiogram you would expect to see.

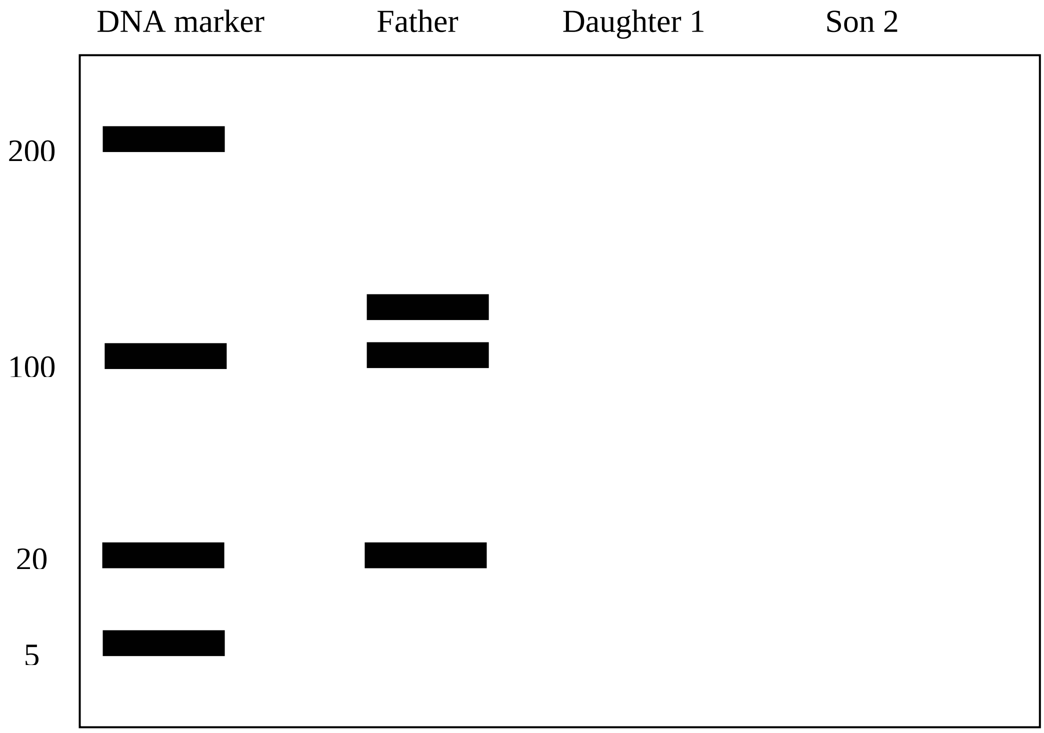


Fig. 2.2

[2]

(ii) Explain how a probe hybridizes to a DNA fragment.

____

____ [2]

(iii) Explain how DNA fragments can be separated into its bands.

____

____ [2]

(iv) Explain how differential banding patterns can result when chromosome 20 is isolated from the father, daughter 1, son 2 and cleaved with the same restriction enzyme, *EcoRI*?

____

____

____ [2]

Total: 15 marks

QUESTION THREE

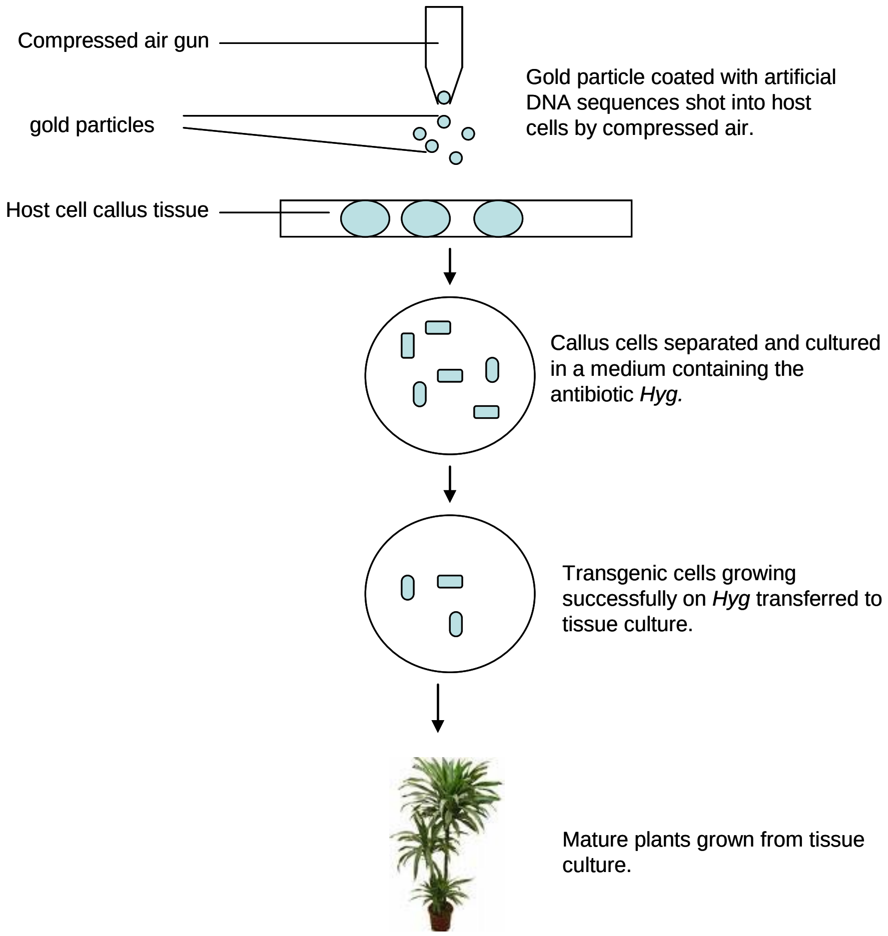
Rice is the staple diet in many parts of the world. It lacks a number of important nutrients, including the ß carotene, from night blindness. Adequate concentrations of vitamin A give protection from night blindness. Higher concentrations act as an antioxidant that may give some protection from cancer and heart disease.
Golden rice, which contains ß carotene, was developed in Switzerland by genetically modifying rice using genes from a daffodil (a flowering plant) and a bacterium.

Figure 3.1 shows an artificial DNA sequence used.

pro	daffodil gene	ter		pro	Hyg resistance	ter

Legend
pro – promoter sequence
ter – terminator sequence

Fig 3.2 shows the main events in obtaining a transgenic plant.



(a) With reference to figures 3.1 and 3.2,

(i) Outline how the daffodil gene might have been isolated from the donor organisms.

____

____

____

____ [2]

(ii) Explain what is meant by :

Transgenic plant:

____ [1]

(iii) Explain the role of *Hyg* resistance gene in this procedure.

____

____

____ [2]

(iv) Another method of producing a transgenic plant is by the use of bacterium named *Agrobacterium*.

Using golden rice as an example or any other named example, outline the process of using *Agrobacterium* to produce a genetically modified crop to improve the yield/nutrition of a plant.

____

____

____

____

____

____ [4]

(b) Tomato plants were transformed by adding an antisense gene *to pTOM13* to the normal genome. Antisense genes, once introduced, are inherited.
Ethene production during fruit ripening was measured in normal plants and plants with one antisense gene. The results are shown in Fig. 3.3

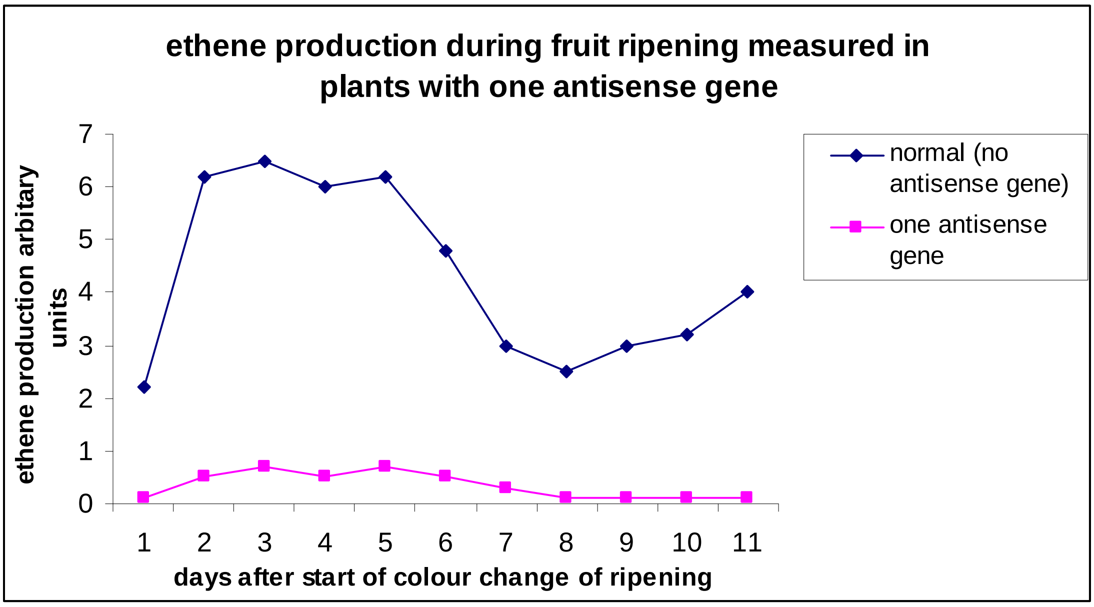


Fig. 3.3

(i) With reference to Fig 3.3,
Comment on the effect on ethene production in the transgenic plant with the antisense gene.

____

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____

____

____ [3]

(ii) Suggest a function of product of *pTOM13* gene in the normal plant.

____

____ [1]

(iii) Describe briefly how the introduction of an antisense gene into the tomato plant reduces the production of ethene in the ripening process.

____

____

____

____ [2]

Total: 16 marks

Write your answers to this question on the separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections **(a), (b)** etc., as indicated in the question.

QUESTION 4

(a) Describe features of a plasmid that made it suitable as a vector. You may want to use a diagram for illustration. [4]

(b) Using a named example, outline the process of cloning a eukaryotic gene into *E.coli* cells to produce eukaryotic protein in genetic engineering. [10]

(c) Discuss the social and ethical implications of production of genetically modified organisms to solve the demand for food in the world. [6]